

Question			Marking details	Marks available																									
				AO1	AO2	AO3	Total	Maths	Prac																				
6	(a)		Indicative response: Before jumping, all of the energy is in the form of gravitational potential energy (GPE). As the jumper falls, the original GPE is transferred to KE. When the cord starts to stretch the KE continues to increase initially, as does the elastic potential energy (EPE). Both GPE and KE will decrease as the cord stretches further, until eventually the KE becomes (momentarily) zero at the lowest point of the jump. At this point the decrease in GPE is equal to the EPE in the cord. As the jumper starts to move upwards the EPE is transferred back to KE and GPE. The jumper does not reach the initial height above the ground, as some energy is transferred to air molecules (and / or reference to hysteresis). Additional comments: Some energy is transferred to air molecules. Explanation of why jumper does not reach original height after first bounce. N.B. Table below is to aid marking – candidates must write in extended prose to gain 6 marks.	2	4		6																						
			<table><tr><th>Position</th><th>E_g</th><th>E_k</th><th>E_{el}</th></tr><tr><td>At start (before jump)</td><td>Max or mgh where h is height above ground</td><td>0</td><td>0</td></tr><tr><td>Free fall, cord slack</td><td>Decreasing</td><td>Increasing</td><td>0</td></tr><tr><td>Cord stretching</td><td>Decreasing</td><td>Continues to increase initially before decreasing.</td><td>Increasing</td></tr><tr><td>At bottom of first bounce</td><td>Minimum - don't accept zero unless explained. Accept $-mg\Delta x$ where Δx is distance fallen</td><td>0</td><td>Maximum or $\Delta E_{el} = -mg\Delta x$</td></tr></table>	Position	E_g	E_k	E_{el}	At start (before jump)	Max or mgh where h is height above ground	0	0	Free fall, cord slack	Decreasing	Increasing	0	Cord stretching	Decreasing	Continues to increase initially before decreasing.	Increasing	At bottom of first bounce	Minimum - don't accept zero unless explained. Accept $-mg\Delta x$ where Δx is distance fallen	0	Maximum or $\Delta E_{el} = -mg\Delta x$						
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			5-6 marks Comprehensive analysis of all 3 energy forms. Reference to energy loss due to air resistance and / or hysteresis. <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured.</i> 3-4 marks Comprehensive account of two energy forms or a limited account of all three. No reference to energy losses. <i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure.</i> 1-2 marks Limited account of one or two energy forms. No reference to energy losses. <i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure.</i> 0 marks <i>No attempt made or no response worthy of credit.</i>						
6	(b)	(i)	Returns to original shape (unstretched length)	1			1		1
		(ii)	[Work done to] straighten (untangle / unravel) polymer chains / molecules Accept C-C bonds are rotated	1			1		1

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		(iii)	Valid method for calculating the area (1) Correct value based on valid method (expect 6.0 ± 0.4 [J]) (1) Valid methods are: Counting squares expect 6 large squares, expect 24 of the $\frac{1}{4}$ squares and expect 600 tiny squares OR Trapezium (0,0), (0.3, 0), (0.3, 30), (0.2, 30) Note that this is equivalent to a triangle and a quadrilateral. Expect: $\frac{1}{2} \times (0.1 + 0.3) \times 30$ OR $\frac{1}{2} \times 0.2 \times 30 + 0.1 \times 30$ OR Trapezium (0,0), (0.3, 0), (0, 10), (0.2, 30) Note that this is equivalent to a triangle and a quadrilateral. Expect very similar numbers: $\frac{1}{2} \times (10 + 30) \times 0.3$ OR $\frac{1}{2} \times 20 \times 0.3 + 10 \times 0.3$ OR Triangle (0,0), (0.3, 0), (0.3, 40) Expect the numbers: $\frac{1}{2} \times 0.3 \times 40$ OR Triangle and trapezium method. Expect the numbers: $\frac{1}{2} \times 0.1 \times 20 + \frac{1}{2} \times (20 + 30) \times 0.2$ (= 1 J + 5 J) OR Anything else which is clearly correct award the marks e.g. triangle-trapezium-trapezium all with a base of 0.1 gives 6.1 [J] The final mark is for the correct answer 6.0 ± 0.4 [J]		2		2	2	2
		(iv)	Hysteresis (1) Difference between energy stored in the rubber band when it is stretched and [useful] energy recovered from it when it is unstretched. Or energy dissipated (or described) [lost] in one cycle of loading and unloading (1)	2			2		2
			Question 6 total	6	6	0	12	2	6